

→* Chapter → 1 *

→* Introduction to Programming *

① Introduction of Computer System →

→ A Computer system is an electronic device that accept data or input process is according to instruction, store data and result process useful information as output.

② Basic Components of Computer System →

→ A Computer system mainly consist of the following components like - input, processing, storage and output.

INPUT PROCESSING STORAGE OUTPUT

a) INPUT → Input unit is used to enter data and instruction into the computer
Keyboard, mouse, scanner, joystick etc.

(*) C.P.U → It is a called the brain of computer. it perform calculation, process instructions and control other component.

The C.P.U mainly contain →

a) A.L.U → Arithmetic Logic Unit perform addition, subtraction, multiplication, Division, Logic operators such as AND, OR, NOT gate.

b) CU → Control units perform fetching instructions, decode instructions, control executing, co-ordinate different part of computer.

* Register → its are small high speed storage location inside the C.P.U used to temporary hold the data and instructions.

* Memory Unit → Memory store program data and intermediate final result.

a) Primary memory → RAM → are temporary and volatile memory.

ROM → Non volatile memory containing permanent instructions.

* Secondary Unit →

Secondary unit perform store data permanent

Hard disk, SSD card, pen drive, memory card etc

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⑧ Output Unit → It provide the process result to the user like monitor, printer, speaker, projector.

⑧ Software → Software is the collection of program and instruction that's tell what to do the computer.

→ Types of Software ←

- i) System Software
- ii) Operating Software
- iii) driver Software
- iv) Compiler Software
- v) Device Software

⑧ Application Software → It is used to user like ms word, Browser, media player.

→* Algorithm and flowchart *←

Algorithm →

Algorithm is a typical programming class can be divided into two phases.

1) First, problem solving phase.

Produce an order sequence of steps that describe solution of the problem. This sequence of steps is called algorithm.

2) Second, the implement phase.

The implement phase of a program in some of program language like C program, Python, etc.

The algorithm successful request to get step by step detail algorithm that is very close to a computer language.

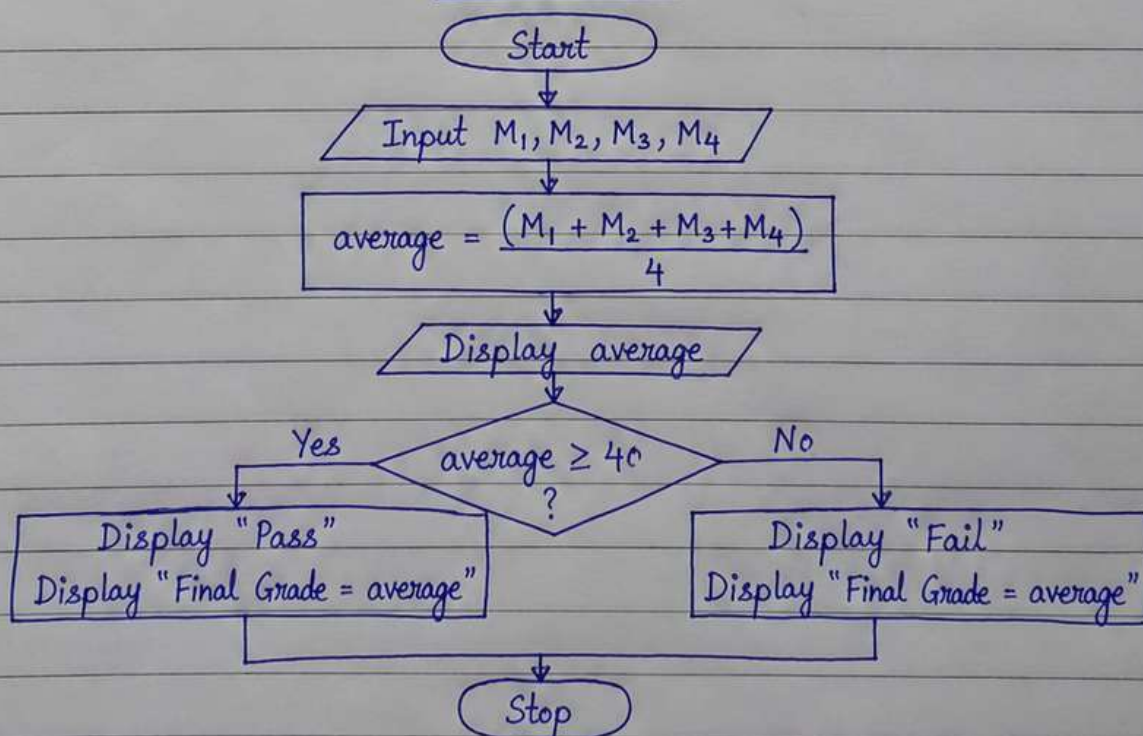
Ques:- Write an algorithm to determine the student final grade and indicate whether it is passing or a failing the final grade separated as a average of four marks.

Solution:-

Algorithm:-

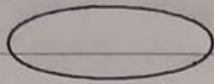
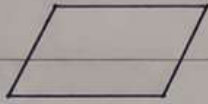
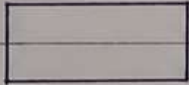
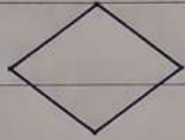
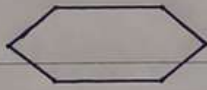
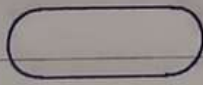
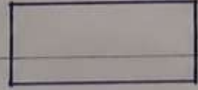
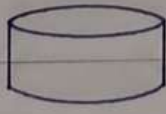
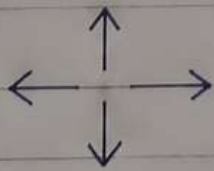
- 1) Start
- 2) Input four subject marks : M_1, M_2, M_3, M_4 .
- 3) Calculate average = $\frac{M_1 + M_2 + M_3 + M_4}{4}$
- 4) Display average.
- 5) If average ≥ 40 then
 Display "Pass" and "Final Grade = average"
 Else
 Display "Fail" and "Final Grade = average"
- 6) Stop.

Flowchart



The flowchart is a graphical representation of sequence of operation in an information system or program. Program show the sequence of instruction in a single program of different symbol as used to draw each type of flowchart.

Flowchart Symbols

	<u>Name</u>	<u>Symbol</u>	<u>Use in flowchart</u>
(i)	Oval		Denotes the beginning or end of the program.
(ii)	Parallelogram		Denotes an input operation or output operation.
(iii)	Rectangle		Denotes a process to be carried out. eg:- Add, Sum, decision etc.
(iv)	Diamond		Denotes a decision or branch to be made the program. Should contain along one or two routes (eg:- IF/THEN/ELSE)
(v)	Hybrid		Denotes an output operation.
(vii)	Flowline		Denotes the direction of logic flow in the program.
(viii)	Terminal		Indicates the beginning and end point of an module.
(viii)	Process		Shows an instruction other than input/output or selection.
(ix)	Disk Storage		Indicate input from or output to the disk storage.
(x)	flow line		indicate the logical sequence of execution steps in the Algorithm.

Question number:- 1

Create algorithm and flowchart that will be output for greatest common factor, (GCF).

Question number:- 2

Create an algorithm and flowchart that will be output for Fibonacci series of two given number.

Question number :- 3

Create an algorithm and flowchart that will be output for all prime number between two numbers.